Working Paper # 20

Revision of BE Software Engineering Program

Sponsored by SEECS

Background

1. The aim of this working paper is to make the BESE curriculum in line with PEC Engineering Education Guidelines issued for the adoption of HEC UEP-2023 (EC letter no. 15-54/Coord/2019/HEC/(CURR/UEP)/578 dated July 13, 2023.

Rationale

2. PEC issued guidelines (PEC letter no. PEC/EAD/P&C/UEP/2023 dated September 27, 2023) for incorporation of HEC UEP-2023 in undergraduate education programs. Revision of the SE curriculum is proposed for compliance with the PEC Framework.

Educational Objectives

3. SE curriculum compliance with PEC guidelines incorporating HEC UEP-2023.

Proposed Changes

4. As per PEC's proposed guideline

   a. Two English courses composed of 3 Cr Hrs each are required in Knowledge Profile WK7 (Engineering in Society). Table 1 shows the courses suggested by PEC.

   b. The standard range prescribed to qualify for the undergraduate engineering degree is 130-136 Cr Hrs with a normal range of 15-18 Cr Hrs in each semester. Currently, 3 courses of 2 Cr Hrs each are offered in semesters 1, 2, and 5. Introducing two courses of 3 Cr Hrs will exceed the limit of 18 Cr Hrs in one semester. The approved curriculum currently offered is shown against the proposed SE Curriculum in Annex A. The proposed curriculum has credit hours within the limit of 18 for each semester.

   c. Ideology and Constitution of Pakistan of 2 Cr Hrs will be offered instead of Pakistan Studies of 2 Cr Hrs in the same semester.

d. The course of Ethics of 2 Cr Hrs will be offered alongside Islamic Studies for non-muslim students.

Table 1: Revised Courses of PEC Framework-2023

Knowledge Profile	Knowledge Area	Sub- Area	Courses	Credit hours
WK-7 (Engineering in Society)	Humanities	English	Functional English	3
			Expository Writing	3
		Culture	Islamic Studies/Ethics	2
			Ideology and Constitution of Pakistan	2

The summary of the proposed changes is listed in Table 2

Table 2: Summary of Proposed Changes

Sr. No	Change
1	Functional English (3-0) will be taught instead of English (2-0) in 1st Semester
2	Expository Writing (3-0) will be taught instead of Communication Skills (2-0) in 2nd Semester
3	Occupational Health and Safety (1-0) will be moved to 1st to 2nd Semester
4	Islamic studies (2-0) will be moved from 2nd semester to 4th Semester
5	Ethics (2-0) will be offered along with Islamic Studies (2-0) for non-Muslim students in 4th semester
6	Sociology elective-1 will be moved from the 4th semester to the 5th semester
7	Technical and Business Writing (2-0) will be removed from 5th Semester
8	Pakistan Studies (2-0) will be replaced with Ideology and Constitution of Pakistan(2-0) in 1st semester

Approval by DBS/FBS

5. Approved by FBS and minutes of FBS attached at Annex B.

Comments of Academics Directorate

6. The proposal was deliberated and endorsed by UCRC held on 23 July 2024.

Recommendation of Academics Directorate

7. Revision of BE Software program at SEECS and MCS being a unified program is recommended for approval w.e.f Fall 2024 batch.

8. Academic council is requested for the decision.

Prepared by: Ms. Mehwish Kiran, Professor Dr. Rafia Mumtaz (HoD SE)

Date: 10th June 10, 2024

Checked by Dr. Muhammad Ajmal Khan (Principal SEECS)

Date_____

Approved by _____

Date_____

Annex-A

Approved Scheme in 66th ACM				Proposed Curriculum			
Semester I				**Semester I**			
Course Code	**Subjects**	**Credit Hrs**		**Course Code**	**Subjects**	**Credit Hrs**	
HU-100	English	2		HU-114	Functional English*	3	
CS-110	Fundamentals of Computer Programming	3-1		CS-110	Fundamentals of Computer Programming	3-1	
HU-107	Pakistan Studies	2		HU-XXX	Ideology and Constitution of Pakistan*	2	
MATH- 101	Calculus and Analytical Geometry	3		MATH-101	Calculus and Analytical Geometry	3	
PHY-102	Applied Physics	2-1		PHY-102	Applied Physics	2-1	
MATH- 161	Discrete Mathematics	3		MATH-161	Discrete Mathematics	3	
OHS-101	Occupational Health and Safety	1					
	Total CHs	**16-2 (18)**			**Total CHs**	**16-2 (18)**	
Semester II				**Semester II**			
CS-212	Object Oriented Programming (OOP)	3-1		CS-212	Object Oriented Programming (OOP)	3-1	
HU-101	Islamic Studies	2		OHS-101	Occupational Health and Safety	1	
MATH-121	Linear Algebra andODEs	3		MATH-121	Linear Algebra andODEs	3	
ME-109	Engineering Drawing	0-2		ME-109	Engineering Drawing	0-2	
HU-109	Communication Skills	2-0		HU-XXX	Expository Writing*	3	
EE-122	Computer Architecture and Logic Design	3-1		EE-122	Computer Architecture and Logic Design	3-1	
ME-105	Workshop Practice	0-1		ME-105	Workshop Practice	0-1	
	Total CHs	**13-5 (18)**			**Total CHs**	**13-5 (18)**	
Semester III				**Semester III**			

CS-220	Database Systems	3-1		CS-220	Database Systems	3-1	
SE-200	Software Engineering	3		SE-200	Software Engineering	3	
CS-250	Data Structures & Algorithms	3-1		CS-250	Data Structures & Algorithms	3-1	
MATH-361	Probability and Statistics	3		MATH-361	Probability and Statistics	3	
EE-353	Computer Networks	3-1		EE-353	Computer Networks	3-1	
	Total CHs	**15-3 (18)**			**Total CHs**	**15-3 (18)**	
Semester IV				**Semester IV**			
CS-330	Operating Systems	3-1		CS-330	Operating Systems	3-1	
CS-251	Design & Analysis of Algorithm	3		CS-251	Design & Analysis of Algorithm	3	
MATH-232	Complex Variables and Transform	3		MATH-232	Complex Variables and Transform	3	
SE-211	Software Design and Architecture	2-1		SE-211	Software Design and Architecture	2-1	
MATH-352	Numerical Methods	2-1		MATH-352	Numerical Methods	2-1	
	Sociology Elective – I	2		HU-101/ HU-XXX	Islamic Studies/ Ethics*	2	
	Total CHs	**15-3 (18)**			**Total CHs**	**15-3 (18)**	
Semester V				**Semester V**			
	Management Science Elective – I	3			Management Science Elective – I	3	
	Engineering Elective –I	3-1			Engineering Elective –I	3-1	
SE-315	Cloud Computing	2-1		SE-315	Cloud Computing	2-1	
	MDEE – I	2-1			MDEE – I	2-1	
HU-212	Technical & Business Writing	2			Sociology Elective – I	2	
SE-314	Software Construction	2-1		SE-314	Software Construction	2-1	
	Total CHs	**14-4 (18)**			**Total CHs**	**14-4 (18)**	
Semester VI				**Semester VI**			
SE-320	Formal Methods	3		SE-320	Formal Methods	3	
SE-321	Software Quality Engineering	3		SE-321	Software Quality Engineering	3	

	Management Science Elective – II	2			Management Science Elective – II	2	
	Engineering Elective-II	3-X			Engineering Elective-II	3-X	
SE-430	Software Project Management	3		SE-430	Software Project Management	3	
CS-261	Human Computer Interaction	2-1		CS-261	Human Computer Interaction	2-1	
	Total CHs	**16-(1+X) (17+X)**			**Total CHs**	**16-(1+X) (17+X)**	
Semester VII				**Semester VII**			
SE-498	Final Year Design Project – I	0-2		SE-498	Final Year Design Project – I	0-2	
	Engineering Elective –III	3-X			Engineering Elective –III	3-X	
	MDEE – II	3			MDEE – II	3	
	Sociology Elective – II	2			Sociology Elective – II	2	
CS-484	Information Security	3		CS-484	Information Security	3	
	Total CHs	**11-(2+X) (13+X)**			**Total CHs**	**11-(2+X) (13+X)**	
Semester VIII				**Semester VIII**			
SE-499	Final Year Design Project – II	0-4		SE-499	Final Year Design Project – II	0-4	
	Engineering Elective –IV	3-1			Engineering Elective –IV	3-1	
CCE-401	Civic and Community Engagement	1-1		CCE-401	Civic and Community Engagement	1-1	
	Engineering Elective –V	3-1			Engineering Elective –V	3-1	
	Total CHs	**7-7 (14)**			**Total CHs**	**7-7 (14)**	
	Grand Total	**107+27+X (134+X)**			**Grand Total**	**107+27+X (134+X)**	

*New Courses to be introduced as per PEC Guidelines